Unit 10: Digital Preservation, Security and Continuity

- Digital preservation is ongoing management of digital objects so their meaning, authenticity and usability survive change.
- A stored file may become unreadable, corrupt, inaccessible, context-free or unlawfully disclosed; preservation addresses all these risks.
- BLISS protects digital learning collections; BRIM safeguards trustworthy institutional records, provenance and restricted content.

TAKEAWAY **Preservation is a managed service, not a single storage purchase.**

Learning outcomes and applied evidence

- Explain preservation, availability, authenticity, integrity, security, and continuity using appropriate distinctions.
- Design copies, fixity checks, permissions, migration decisions and restoration testing for a small fictional collection.
- Produce a preservation plan with accountable staff, risk priorities, records and recovery evidence.

TAKEAWAY A good plan is testable, documented and appropriate to the collection.

Why digital information disappears

- Storage media can fail or be lost; an incorrect deletion or malicious change may affect multiple synchronised copies.
- Obsolete software or formats may prevent future rendering even when every byte remains unchanged.
- Missing metadata, rights information or staff knowledge can make an intact object unusable.

TAKEAWAY Bit-level survival alone does not guarantee access or trust.

Digital object: beyond the file

- An object may include original bits, technical and descriptive metadata, a transcript, relationships and rights information.
- Preservation packages maintain these parts together so an item can still be interpreted years later.
- For a scanned minute book, the image, page order, identifier, capture events and access rules all matter.

TAKEAWAY **Maintain context as well as content.**

Preservation lifecycle

- Select and appraise -> receive -> verify -> register -> store -> monitor -> manage change -> provide controlled access.
- Capture provenance and preservation events whenever an object is ingested, copied, repaired or migrated.
- Policies set responsibilities and review periods; operational evidence demonstrates that steps occurred.

TAKEAWAY **Preservation work starts at acquisition, not after an incident.**

Preservation master and access copy

- A preservation master retains the most appropriate source representation and is changed only under controlled procedures.
- An access derivative can be resized, compressed, watermarked or redacted for permitted users.
- Never substitute an edited public copy for the only retained original; document links between versions.

TAKEAWAY Preservation and delivery have different requirements.

What does a checksum tell us?

- A cryptographic digest such as SHA-256 provides a reproducible fingerprint of a byte sequence.
- Comparing a stored trusted baseline digest with a freshly computed digest helps detect unexpected byte changes.
- A matching checksum does not prove who created the content, that the source was legitimate or that access is authorised.

TAKEAWAY Fixity detects change; provenance and authentication answer different questions.

Example: a fixity manifest

- A manifest lists stable file identifiers, paths, expected SHA-256 values, timestamps and checks performed.
- Example: record_001.tif -> expected 4e5f...; a comparison result of 4e5f... indicates the bytes match the baseline.
- A mismatch triggers investigation, containment and recovery from a verified copy; do not silently overwrite evidence.

TAKEAWAY Record both successful checks and exceptions in an audit log.

Establishing a trustworthy baseline

- Generate hashes when files are received or created; validate that files, names, quantities and associated documentation are complete.
- Store the baseline manifest in an access-controlled location and preserve it separately from the files it describes.
- If a received file is already corrupt, computing a perfect checksum will faithfully record that bad state.

TAKEAWAY The baseline must be connected to an accountable ingest process.

Storage copies: the 3-2-1 planning heuristic

- Plan at least three copies, use at least two storage types or sufficiently independent systems, and keep one copy off-site.
- Geographic separation reduces common-disaster exposure; independent credentials or an offline/immutable copy can reduce ransomware risk.
- 3-2-1 is a starting mnemonic, not a certification: copies need checks, inventories and successful restoration tests.

TAKEAWAY **Copy diversity and recoverability matter more than a slogan.**

A practical three-copy design

- Primary: controlled repository storage for approved current use and scheduled integrity monitoring.
- Secondary: independent local backup with restricted write access; tertiary: encrypted off-site copy under separate recovery controls.
- Keep retention rules, ownership, encryption keys, monitoring and test-restoration logs for each copy.

TAKEAWAY A backup without readable keys and a verified restore path may be unusable.

Backup, replication and archival storage

- A backup is a recoverable historical copy; replication mirrors data and can reproduce accidental deletion or corruption quickly.
- Archival storage prioritises durable retention and controlled retrieval; it is not automatically a usable backup of production systems.
- Snapshots, replication and backups can complement one another but have different failure modes.

TAKEAWAY Define each copy by the problem it solves.

Recovery point objective (RPO)

- RPO is the maximum tolerable age of data recovered after a disruption; it is expressed as an amount of time.
- Hourly backups might leave up to nearly one hour of changes unrecovered, if the backups work as intended.
- For frequently updated records, determine an acceptable RPO before deciding on backup frequency.

TAKEAWAY Backup schedule should be driven by allowable data loss.

Recovery time objective (RTO)

- RTO is the intended time to restore a service after an outage, including required systems, staff and validation.
- A stored copy may take hours or days to retrieve, decrypt, restore, validate and reconnect to users.
- Agree RTO with stakeholders and test whether the actual recovery procedure can meet it.

TAKEAWAY Having bytes available is not the same as restoring a working service.

Worked continuity case

- A catalogue backs up every 4 hours: worst-case change loss may approach 4 hours if the latest backup is healthy.
- If the tested end-to-end restoration takes 7 hours, an RTO target of 2 hours is not currently achievable.
- Reduce gaps by revising backup frequency, recovery design, staffing or service priorities and retesting.

TAKEAWAY State assumptions and record tested recovery performance.

Integrity, availability and confidentiality

- Integrity means preventing or detecting unauthorised or accidental change; availability means authorised users can access needed services.
- Confidentiality means information is disclosed only to approved people; safeguarding some records also requires accountability.
- A public portal may need high availability, while a confidential personnel archive requires strict access decisions.

TAKEAWAY Security controls must fit the sensitivity and mission of each collection.

Role-based and least-privilege access

- Define responsibilities: public visitor, library staff, records officer, preservation operator and system administrator.
- Grant minimum privileges required for a task; separate access to view public derivatives from rights to alter masters.
- Authentication, approval workflows, audit logs and periodic privilege reviews support accountability.

TAKEAWAY A hidden web link is not an access-control system.

Protecting collections from ransomware

- Apply patch management, multifactor authentication, least privilege, secure configurations and staff awareness.
- Maintain protected/offline/immutable recovery copies where feasible; monitor unusual deletions and mass file changes.
- In an incident, isolate affected systems, preserve evidence and follow the institutional response process.

TAKEAWAY Prevent, detect, respond and recover rather than relying on backups alone.

NIST CSF 2.0: six functions

- Govern sets risk strategy, roles and policy; Identify understands assets, dependencies and exposure.
- Protect implements safeguards; Detect finds suspicious events or anomalies.
- Respond contains and investigates; Recover restores capability and learns from the event.

TAKEAWAY The six functions are concurrent outcomes, not a one-way checklist.

NDSA Levels of Digital Preservation v2.1

- The NDSA Levels provide a staged framework for assessing a digital preservation programme across technical and organisational areas.
- Version 2.1 (March 2026) adds environmental sustainability guidance; the levels support incremental improvement.
- A collection should select suitable targets based on risk, resources and purpose rather than assuming the highest level is always necessary.

TAKEAWAY **Maturity assessment supports prioritisation; it does not replace local policy.**

Preservation metadata and event logs

- Track object IDs, rights, original formats, checksums, storage locations, agents and events.
- A preservation event records what happened, when, why, using what software and with what outcome.
- Keep enough provenance to show how a master became a derivative or a migrated format.

TAKEAWAY **Document decisions so future custodians can interpret evidence.**

Format obsolescence and monitoring

- Risk arises when a format, application, codec, hardware or licence becomes difficult to access or support.
- Monitor inventories and dependencies; document software required to render less common or interactive objects.
- An open or well-documented format may lower risk, but formats still need assessment and testing.

TAKEAWAY **File extension alone does not determine future preservation risk.**

Format migration: a controlled intervention

- Migration transforms an object into another format for continued use, and may change layout, features or hidden data.
- Retain the original where appropriate; preserve metadata and records of transformation, validation and approval.
- Compare renderings and significant properties, not only checksums: changed-format files are expected to have different hashes.

TAKEAWAY A successful migration is evidenced by functional and content checks.

Emulation and complex multimedia

- Emulation reproduces elements of an earlier hardware or software environment so interactive resources can be experienced.
- It may be more suitable than flat export for historic websites, interactive CDs or software-dependent exhibits.
- Emulation also requires licensing, security, environment documentation and ongoing maintenance.

TAKEAWAY Preserve meaningful behaviour when static conversion would lose it.

Security and preservation can conflict

- An original record may need strict retention while users need a redacted access copy.
- Tight encryption protects confidentiality but lost recovery keys can permanently prevent access.
- Retention laws, privacy, authenticity and access must be balanced through documented institutional governance.

TAKEAWAY Security must enable authorised long-term use, not accidentally destroy it.

Threat and risk register

- Describe each asset, hazard, likelihood, impact, affected users, owner, present controls and next action.
- A simple likelihood x impact score can help classroom prioritisation, but professional risk methods also consider uncertainty.
- Review risks after system changes, new laws, incidents and evolving collection needs.

TAKEAWAY The most useful risk record identifies an owner and a feasible response.

Business continuity versus disaster recovery

- Business continuity considers how essential services can continue during disruption, including temporary alternatives.
- Disaster recovery restores affected technology, data and infrastructure after disruption.
- For a library, continuity may include offline circulation; for records, a controlled paper register may support essential operations.

TAKEAWAY Continuity keeps critical work going while recovery repairs underlying systems.

A practical incident response sequence

- Recognise and report unusual activity; involve the appointed incident lead and follow the organisation’s procedure.
- Contain damage and preserve logs/evidence; assess scope and confirm the safety of candidate restore points.
- Restore in priority order, validate integrity and access, document lessons and update controls.

TAKEAWAY Do not conceal an incident or restore compromised files without validation.

Test restores and evidence

- A restoration test retrieves a selected backup, checks integrity and confirms the material can actually be opened.
- A service-level recovery exercise also tests permissions, metadata links, dependencies, staff roles and communication.
- Record test date, backup version, time taken, failures, corrective actions and next test.

TAKEAWAY A successful backup job is not proof of successful restoration.

Collection case: BLISS digital library

- A digital exhibition stores preservation TIFF masters, smaller JPEG access images, captions, catalogue metadata and rights notices.
- Plan separate backups, integrity checks, consistent identifiers, public portal availability and staff restore responsibilities.
- If the access site fails, users may temporarily receive an approved index while the service is recovered.

TAKEAWAY Preservation and public access should remain separate but linked.

Collection case: BRIM records archive

- A digitised committee series has source masters, page-order relationships, restriction decisions and event history.
- Access to some records requires named officers; checksums and preservation logs support custody and continuity.
- A restoration must recover the correct records together with permissions, provenance and audit history.

TAKEAWAY **Restoring files without their context can compromise record trustworthiness.**

Environmental sustainability

- Retention should follow documented appraisal, legal obligations and value rather than copying everything indefinitely.
- Consider energy, hardware lifetime, storage tiering and verification frequency while maintaining suitable controls.
- NDSA v2.1 highlights sustainability as one consideration in preservation decisions, not permission to neglect risk.

TAKEAWAY Sustainable stewardship balances preservation needs and resource impacts.

Guided practical: preserve a fictional collection

- Inspect the supplied four-file manifest, identify one simulated integrity exception and document an investigation plan.
- Propose three independently protected copies, a role-permission matrix and realistic RPO/RTO values.
- Complete a test-restoration record, a risk register and a brief BLISS or BRIM access workflow.

TAKEAWAY Students submit evidence of decisions, not confidential real records.

Common misconceptions

- A checksum match does not confirm the file’s authenticity or prove that its content is safe to publish.
- Three synchronised replicas are not automatically three independent restorable backups.
- Changing format alters bytes; compare significant properties and document migration rather than expecting matching hashes.

TAKEAWAY Correct terminology improves operational decisions.

Review, reflection and professional judgement

- What needs to survive: individual bytes, visible appearance, ability to search, interactive behaviour, evidential value or all of these?
- What could interrupt continuity: deletion, fire, format loss, loss of keys, ransomware or lack of staff knowledge?
- Which controls would a small resource-limited information service prioritise first, and how would success be demonstrated?

TAKEAWAY Select safeguards based on real use, risk, resources and accountability.

Unit 10: key ideas

- Inventory and describe collections; preserve content with independent copies, fixity checks and tested recovery.
- Protect sensitive materials through authorised access; manage changes such as migration using preservation metadata and validation.
- Preservation decisions also require legal, ethical and inclusive access practices.

TAKEAWAY Preservation is a continuing institutional responsibility supported by evidence.